Minimax lower bound for Sobolev nonparametric regression

1 Preliminaries

Notation. We write $[N] := \{1, \dots, N\}$. Universal positive constants are denoted by $c, C, c_0, C_1, \dots$, possibly depending on s, d, σ, and a fixed bump function ψ to be defined below; their values may change from line to line. The dependence on s, d, σ, ψ is suppressed in $\lesssim$, $\gtrsim$, and $\asymp$.

For a multi-index $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_{\geq 0}^d$, let $|\alpha| = \sum_i \alpha_i$ and $\partial^\alpha = \partial_{x_1}^{\alpha_1} \cdots \partial_{x_d}^{\alpha_d}$. For $s \in \mathbb{N}_{\geq 1}$, the Sobolev seminorm and norm on $[0,1]^d$ are

$$\|f\|_{W_2^s}^2 \;:=\; \sum_{|\alpha| \leq s} \int_{[0,1]^d} \big(\partial^\alpha f(x)\big)^2 \, dx.$$

We write $\|f\|_{L^2}^2 := \int_{[0,1]^d} f(x)^2 \, dx$ and $\|f\|_{L^\infty} := \sup_{x \in [0,1]^d} |f(x)|$.

The *empirical design norm* on a fixed sample $(x_1, \dots, x_n) \in ([0,1]^d)^n$ is

$$\|h\|_n^2 \;:=\; \frac{1}{n} \sum_{i=1}^n h(x_i)^2.$$

Definition 1.1 (Sobolev unit ball)**.** The unit ball of the Sobolev space $W_2^s([0,1]^d)$ is

$$\mathcal{F} \;:=\; \big\{ f : [0,1]^d \to \mathbb{R} \;:\; f \in W_2^s([0,1]^d),\ \|f\|_{W_2^s} \leq 1 \big\}.$$

Assumption 1.2 (Fixed design and Gaussian noise)**.** The smoothness index $s \in \mathbb{N}_{\geq 1}$ and dimension $d \in \mathbb{N}_{\geq 1}$ are fixed. The sample size satisfies $n \geq n_0(s, d)$, where n_0 is a constant defined in the proof. The design $(x_1, \dots, x_n) \in ([0,1]^d)^n$ is fixed and balanced in the sense of Definition 1.4 below. For each $f \in \mathcal{F}$, observations $Y_i = f(x_i) + \varepsilon_i$ are drawn with $\varepsilon_1, \dots, \varepsilon_n \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2)$ and $\sigma > 0$ known.

Remark 1.3. Assumption 1.2 is the standard fixed-design Gaussian regression setup. The balanced-design condition is mild and is satisfied by the regular grid $x_i = ((i_1 - 1/2)/n^{1/d}, \dots)$ on $[0,1]^d$ whenever n is a d-th power; it ensures that each grid cell used in the bump construction contains a controllable number of design points. The Gaussian-noise assumption yields the closed-form KL divergence used in Fact 1.8.

Definition 1.4 (Grid and balanced design)**.** For an integer $m \geq 1$, partition $[0,1]^d$ into m^d disjoint half-open cells

$$R_k \;:=\; \prod_{\ell=1}^d \Big[\tfrac{k_\ell - 1}{m}, \tfrac{k_\ell}{m} \Big), \qquad k = (k_1, \dots, k_d) \in [m]^d,$$

of equal volume m^{-d} (boundary points assigned to the cell with smaller index). The design $(x_1, \dots, x_n)$ is *balanced at scale* m if every cell R_k contains at most $C_{\mathsf{bal}} \cdot n/m^d$ design points, for a universal constant $C_{\mathsf{bal}} \geq 1$.

Remark 1.5. The regular grid x_i described in Remark 1.3 is balanced at every scale m that divides $n^{1/d}$; one may take $C_{\mathsf{bal}} = 1$ in that case. We state Definition 1.4 more generally so that the lower bound also covers approximately uniform designs.

Definition 1.6 (Risk and minimax risk)**.** The L^2-risk of an estimator $\widehat{f} = \widehat{f}(\mathcal{D}_n)$ at $f^* \in \mathcal{F}$ is

$$R(\widehat{f}, f^*) \;:=\; \mathbb{E}_{\mathcal{D}_n}\big[\, \big\|\widehat{f} - f^*\big\|_{L^2}^2 \,\big],$$

where the expectation is taken over the noise $(\varepsilon_1, \ldots, \varepsilon_n)$. The minimax risk over $\mathcal{F}$ is

$$\mathfrak{M}_n \;:=\; \inf_{\widehat{f}} \sup_{f^* \in \mathcal{F}} R(\widehat{f}, f^*),$$

with the infimum over all measurable estimators $\widehat{f} : (\mathbb{R}^n) \to L^2([0,1]^d)$.

Fact 1.7 (Restatement of Varshamov–Gilbert, [1])**.** For every integer $K \geq 8$, there exists a subset $\{\omega^{(0)}, \omega^{(1)}, \ldots, \omega^{(M)}\} \subset \{0,1\}^K$ with $\omega^{(0)} = (0, \ldots, 0)$ such that

$$M \;\geq\; 2^{K/8} \qquad \text{and} \qquad \rho_H(\omega^{(j)}, \omega^{(k)}) \;\geq\; K/8 \quad \forall\, 0 \leq j < k \leq M,$$

where ρ_H denotes Hamming distance.

Fact 1.8 (KL for Gaussian fixed-design regression)**.** Under Assumption 1.2, for two regression functions f, g, the laws P_f, P_g of the observation vector $(Y_1, \ldots, Y_n)$ satisfy

$$\mathrm{KL}(P_f \,\|\, P_g) \;=\; \frac{1}{2\sigma^2} \sum_{i=1}^{n} \big(f(x_i) - g(x_i)\big)^2 \;=\; \frac{n}{2\sigma^2}\, \|f - g\|_n^2 \,.$$

2 Main result

Theorem 2.1 (Minimax lower bound over the Sobolev ball)**.** *Under Assumption 1.2, there exist constants $c_{\mathsf{lb}} = c_{\mathsf{lb}}(s, d, \sigma) > 0$ and $n_0 = n_0(s, d)$ such that for all $n \geq n_0$,*

$$\mathfrak{M}_n \;=\; \inf_{\widehat{f}} \sup_{f^* \in \mathcal{F}} \mathbb{E}_{\mathcal{D}_n}\big[\, \big\|\widehat{f} - f^*\big\|_{L^2}^2 \,\big] \;\geq\; c_{\mathsf{lb}} \cdot n^{-2s/(2s+d)}. \tag{1}$$

In particular, $\mathfrak{M}_n \gtrsim n^{-2s/(2s+d)}$.

Remark 2.2. The rate $n^{-2s/(2s+d)}$ in Theorem 2.1 is the well-known minimax rate for Sobolev regression. The matching upper bound (e.g. via spline / wavelet projection or local polynomial regression) is classical and not pursued here.

3 The bump-function family

We construct a finite family of regression functions in $\mathcal{F}$ that are well separated in L^2 yet close in L^∞. The construction is the standard scaled-bump family at resolution m, indexed by binary strings.

The base bump. Fix once and for all a function $\psi \in C_c^\infty(\mathbb{R}^d)$ supported in the open cube $(0,1)^d$ with $\psi \not\equiv 0$. The choice of ψ does not affect the rate; it only affects the constants c_{lb} and n_0 in Theorem 2.1.

Grid translates. Let $m \geq 2$ be an integer (to be chosen later as $m \asymp n^{1/(2s+d)}$). For $k \in [m]^d$, define

$$\psi_k(x) \;:=\; \psi\big(m \cdot (x - z_k)\big), \qquad z_k := \big((k_1-1)/m, \ldots, (k_d-1)/m\big). \tag{2}$$

Each ψ_k is C^∞ and supported in R_k (Definition 1.4); the supports $\{R_k\}_{k\in[m]^d}$ are pairwise disjoint.

Hypothesis family. For each $\omega \in \{0,1\}^{m^d}$ (indexed by $k \in [m]^d$), define

$$f_\omega(x) \;:=\; \frac{c_\psi}{m^s} \sum_{k\in[m]^d} \omega_k\, \psi_k(x), \qquad c_\psi := \frac{1}{\|\psi\|_{W_2^s}}. \tag{3}$$

Since $\|\psi\|_{W_2^s} > 0$ (because ψ has a non-zero derivative of some order $\leq s$, indeed $\|\psi\|_{L^2} > 0$), c_ψ is well-defined and positive.

The next lemma collects the relevant norms.

Lemma 3.1 (Norms of the bump family)**.** *For every integer $m \geq 2$ and every $\omega, \omega' \in \{0,1\}^{m^d}$:*

(i) $\|f_\omega\|_{W_2^s} \leq 1$, *so* $f_\omega \in \mathcal{F}$.

(ii) $\|f_\omega - f_{\omega'}\|_{L^2}^2 \;=\; c_\psi^2\, m^{-(2s+d)}\; \|\psi\|_{L^2}^2\; \rho_H(\omega,\omega')$.

(iii) $\|f_\omega - f_{\omega'}\|_{L^\infty} \;\leq\; c_\psi\, m^{-s}\; \|\psi\|_{L^\infty}$.

Proof. The proof is a direct change of variables, using that the ψ_k have disjoint supports. Write $g_\psi := \|\psi\|_{W_2^s}$ for brevity.

Part (i). For any multi-index α with $|\alpha| \leq s$ and any $k \in [m]^d$, the chain rule gives

$$\partial^\alpha \psi_k(x) \;=\; m^{|\alpha|} \cdot (\partial^\alpha \psi)\big(m \cdot (x - z_k)\big).$$

By the change of variable $y = m(x - z_k)$ (Jacobian m^{-d}) over R_k,

$$\int_{R_k} \big(\partial^\alpha \psi_k(x)\big)^2\, dx \;=\; m^{2|\alpha|} \int_{[0,1]^d} (\partial^\alpha \psi(y))^2 \cdot m^{-d}\, dy \;=\; m^{2|\alpha|-d}\; \|\partial^\alpha \psi\|_{L^2}^2\,.$$

Because the ψ_k have disjoint supports, the derivatives $\partial^\alpha \psi_k$ are also pairwise disjointly supported, hence

$$\|f_\omega\|_{W_2^s}^2 \;=\; \frac{c_\psi^2}{m^{2s}} \sum_{|\alpha|\leq s} \sum_k \omega_k^2 \int_{R_k} \big(\partial^\alpha \psi_k\big)^2 dx \;=\; \frac{c_\psi^2}{m^{2s}} \sum_k \omega_k^2 \sum_{|\alpha|\leq s} m^{2|\alpha|-d}\, \|\partial^\alpha \psi\|_{L^2}^2\,. \tag{4}$$

For each $|\alpha| \leq s$, $m^{2|\alpha|-d} \leq m^{2s-d}$ since $m \geq 1$. Substituting and using $\omega_k^2 \leq 1$ and $\sum_k 1 = m^d$,

$$\|f_\omega\|_{W_2^s}^2 \;\leq\; \frac{c_\psi^2}{m^{2s}} \cdot m^d \cdot m^{2s-d}\, g_\psi^2 \;=\; c_\psi^2\, g_\psi^2 \;=\; 1,$$

where the last equality uses $c_\psi = 1/g_\psi$. This proves (i).

Part (ii). Set $h := f_\omega - f_{\omega'} = (c_\psi/m^s) \sum_k (\omega_k - \omega'_k)\psi_k$. By disjoint supports,

$$\|h\|_{L^2}^2 \;=\; \frac{c_\psi^2}{m^{2s}} \sum_k (\omega_k - \omega'_k)^2 \int_{R_k} \psi_k(x)^2\, dx \;=\; \frac{c_\psi^2}{m^{2s}} \cdot m^{-d}\; \|\psi\|_{L^2}^2 \sum_k (\omega_k - \omega'_k)^2,$$

where the integral evaluation uses the same change of variable as above with $\alpha = 0$. Since $\omega_k, \omega'_k \in \{0,1\}$, $(\omega_k - \omega'_k)^2 = \mathbf{1}\{\omega_k \neq \omega'_k\}$, so $\sum_k (\omega_k - \omega'_k)^2 = \rho_H(\omega, \omega')$. This proves (ii).

Part (iii). For any $x \in [0,1]^d$, x lies in at most one cell R_k (the cells are disjoint and cover $[0,1]^d$), so the sum $\sum_k \omega_k \psi_k(x)$ has at most one non-zero term, bounded by $\|\psi\|_{L^\infty}$. Hence

$$\|f_\omega - f_{\omega'}\|_{L^\infty} \;\le\; \frac{c_\psi}{m^s} \sup_{x \in [0,1]^d} \Big| \sum_k (\omega_k - \omega'_k)\psi_k(x) \Big| \;\le\; \frac{c_\psi}{m^s} \|\psi\|_{L^\infty},$$

proving (iii) and completing the proof. □

4 KL divergence between hypotheses

Recall from Definition 1.6 that P_f denotes the law of $(Y_1, \ldots, Y_n) = (f(x_i) + \varepsilon_i)_{i=1}^n$ under Assumption 1.2. We bound the KL divergence between P_{f_ω} and $P_{f_{0_{\mathsf{cw}}}}$, where $f_{0_{\mathsf{cw}}} \equiv 0$ corresponds to the zero codeword $0_{\mathsf{cw}} := (0, \ldots, 0) \in \{0,1\}^{m^d}$ (denoted f_0 below).

Lemma 4.1 (KL bound for the bump family)**.** *Suppose the fixed design is balanced at scale m in the sense of Definition 1.4, with constant $C_{\mathsf{bal}} \ge 1$. Then for every $\omega \in \{0,1\}^{m^d}$,*

$$\mathrm{KL}(P_{f_\omega} \,\|\, P_{f_0}) \;\le\; C_{\mathsf{KL}} \cdot \frac{n}{\sigma^2 m^{2s}}, \qquad C_{\mathsf{KL}} := \tfrac{1}{2} c_\psi^2 \, \|\psi\|_{L^\infty}^2 \, C_{\mathsf{bal}}. \tag{5}$$

Proof. The proof has two steps: invoke the closed-form Gaussian KL, then bound the empirical norm of the bump family using the design's balance property.

Step 1 (Gaussian KL). By Fact 1.8 applied to $f = f_\omega$ and $g = f_0 \equiv 0$,

$$\mathrm{KL}(P_{f_\omega} \,\|\, P_{f_0}) \;=\; \frac{1}{2\sigma^2} \sum_{i=1}^n f_\omega(x_i)^2. \tag{6}$$

Step 2 (bounding $\sum_i f_\omega(x_i)^2$). By Eq. (3) and the disjointness of the supports of the ψ_k, partitioning the design points by their cell index gives

$$\sum_{i=1}^n f_\omega(x_i)^2 \;=\; \frac{c_\psi^2}{m^{2s}} \sum_{k \in [m]^d} \omega_k^2 \sum_{i : x_i \in R_k} \psi_k(x_i)^2 \;\le\; \frac{c_\psi^2 \, \|\psi\|_{L^\infty}^2}{m^{2s}} \sum_{k \in [m]^d} \omega_k^2 \cdot |\{i : x_i \in R_k\}|,$$

where the inequality uses $\psi_k(x_i)^2 \le \|\psi\|_{L^\infty}^2$ pointwise. By balance (Definition 1.4), $|\{i : x_i \in R_k\}| \le C_{\mathsf{bal}}\, n/m^d$; combined with $\omega_k^2 \le 1$ and $\sum_k 1 = m^d$,

$$\sum_{i=1}^n f_\omega(x_i)^2 \;\le\; \frac{c_\psi^2 \, \|\psi\|_{L^\infty}^2}{m^{2s}} \cdot m^d \cdot \frac{C_{\mathsf{bal}}\, n}{m^d} \;=\; \frac{C_{\mathsf{bal}}\, c_\psi^2 \, \|\psi\|_{L^\infty}^2 \, n}{m^{2s}}. \tag{7}$$

Step 3 (combine). Substituting Eq. (7) into Eq. (6),

$$\mathrm{KL}(P_{f_\omega} \,\|\, P_{f_0}) \;\le\; \frac{1}{2\sigma^2} \cdot \frac{C_{\mathsf{bal}}\, c_\psi^2 \, \|\psi\|_{L^\infty}^2 \, n}{m^{2s}} \;=\; \frac{C_{\mathsf{KL}} \cdot n}{\sigma^2 m^{2s}},$$

which matches Eq. (5).

This completes the proof. □

Remark 4.2. We state Lemma 4.1 for KL between P_{f_ω} and the reference P_{f_0} because this is the form required by Tsybakov's Fano-type bound (Fact 1.7's companion theorem, used in Section 5). The same bound holds for $\mathrm{KL}(P_{f_\omega} \| P_{f_{\omega'}})$ for any pair ω, ω', by the same argument with $\omega - \omega'$ in place of ω (entries in $\{-1, 0, 1\}$, but the bound $|\omega_k - \omega'_k|^2 \le 1$ still holds).

5 Local Fano application

We instantiate Tsybakov's Fano-type bound on the bump family. The version we use, stated below for convenience, is Theorem 2.5 of [1].

Lemma 5.1 (Tsybakov's Fano-type bound, restated from [1])**.** *Let $(\mathcal{X}, \mathcal{A})$ be a measurable space and $\{P_\theta : \theta \in \Theta\}$ a family of probability measures on it. Let $d(\cdot,\cdot)$ be a semi-metric on Θ. Suppose there exist $\theta_0, \theta_1, \ldots, \theta_M \in \Theta$ with $M \geq 2$ such that:*

(i) $d(\theta_j, \theta_k) \geq 2\rho$ for all $0 \leq j < k \leq M$, with $\rho > 0$;

(ii) $\frac{1}{M}\sum_{j=1}^{M} \mathrm{KL}(P_{\theta_j} \,\|\, P_{\theta_0}) \leq \alpha \log M$ with $0 < \alpha < 1/8$.

Then

$$\inf_{\widehat{\theta}} \sup_{\theta \in \{\theta_0, \ldots, \theta_M\}} P_\theta\big(d(\widehat{\theta}, \theta) \geq \rho\big) \;\geq\; \frac{\sqrt{M}}{1+\sqrt{M}}\left(1 - 2\alpha - \sqrt{\tfrac{2\alpha}{\log M}}\right), \tag{8}$$

where the infimum is over all measurable estimators $\widehat{\theta} : \mathcal{X} \to \Theta$.

Lemma 5.2 (Probability lower bound for the bump family)**.** *Under Assumption 1.2 with balance constant C_{bal}, there exist constants $c_\rho = c_\rho(\psi) > 0$ and $c_{\mathsf{prob}} > 0$ (universal) such that the following holds. Let $m \geq 2$ be an integer with $m^d \geq 8$, and assume*

$$\frac{n}{\sigma^2\, m^{2s+d}} \;\leq\; \alpha_\star, \qquad \alpha_\star \;:=\; \frac{\log 2}{256\, C_{\mathsf{KL}}}. \tag{9}$$

Let $\mathcal{H} \subseteq \{0,1\}^{m^d}$ be the Varshamov–Gilbert packing of Fact 1.7 (with $K = m^d$). Then

$$\inf_{\widehat{f}} \sup_{\omega \in \mathcal{H}} P_{f_\omega}\Big(\Big\|\widehat{f} - f_\omega\Big\|_{L^2} \geq c_\rho\, m^{-s}\Big) \;\geq\; c_{\mathsf{prob}}, \tag{10}$$

where the infimum is over all measurable estimators $\widehat{f} : \mathbb{R}^n \to L^2([0,1]^d)$.

Proof. We apply Lemma 5.1 with $\Theta = \mathcal{H}$, θ_j identified with $\omega^{(j)} \in \mathcal{H}$ (and the corresponding regression function $f_{\omega^{(j)}} \in \mathcal{F}$), and semi-metric $d(\omega, \omega') := \|f_\omega - f_{\omega'}\|_{L^2}$. Set $M + 1 := |\mathcal{H}|$, so by Fact 1.7 (which requires $m^d \geq 8$, satisfied by hypothesis),

$$M \;\geq\; 2^{m^d/8} \;-\; 1. \tag{11}$$

In particular, since $2^{m^d/8} - 1 \geq 2^{m^d/8}/2$ for $m^d \geq 8$, we have $\log M \geq (m^d/8)\log 2 - \log 2 \geq (m^d/16)\log 2$ whenever $m^d \geq 16$; we absorb the finitely many integer cases $8 \leq m^d < 16$ into the n_0 threshold and write $\log M \geq (m^d/16)\log 2$ in what follows.

Verify hypothesis (i) (separation). By Lemma 3.1(ii) and the pairwise-Hamming lower bound in Fact 1.7,

$$\|f_{\omega^{(j)}} - f_{\omega^{(k)}}\|_{L^2}^2 \;=\; c_\psi^2\, \|\psi\|_{L^2}^2\, m^{-(2s+d)}\, \rho_H(\omega^{(j)}, \omega^{(k)}) \;\geq\; \frac{c_\psi^2\, \|\psi\|_{L^2}^2}{8}\, m^{-2s}. \tag{12}$$

Therefore $\|f_{\omega^{(j)}} - f_{\omega^{(k)}}\|_{L^2} \geq 2\rho$ for all $0 \leq j < k \leq M$ with

$$\rho \;:=\; c_\rho\, m^{-s}, \qquad c_\rho \;:=\; \frac{c_\psi\, \|\psi\|_{L^2}}{2\sqrt{8}} \;>\; 0. \tag{13}$$

Verify hypothesis (ii) (KL). By Lemma 4.1 and Eq. (9),

$$\mathrm{KL}\big(P_{f_{\omega(j)}} \,\|\, P_{f_{\omega(0)}}\big) \;\le\; \frac{C_{\mathsf{KL}}\, n}{\sigma^2\, m^{2s}} \;=\; C_{\mathsf{KL}} \cdot m^d \cdot \frac{n}{\sigma^2\, m^{2s+d}} \;\le\; \alpha_\star\, C_{\mathsf{KL}}\, m^d.$$

Averaging over $j \in [M]$ does not change this uniform bound. Using $\log M \ge (m^d/16)\log 2$ from above and the definition $\alpha_\star = (\log 2)/(256\, C_{\mathsf{KL}})$,

$$\frac{1}{M}\sum_{j=1}^{M} \mathrm{KL}\big(P_{f_{\omega(j)}} \,\|\, P_{f_{\omega(0)}}\big) \;\le\; \alpha_\star\, C_{\mathsf{KL}}\, m^d \;\le\; \alpha_\star\, C_{\mathsf{KL}} \cdot \frac{16}{\log 2}\, \log M \;=\; \frac{1}{16}\, \log M. \tag{14}$$

Thus the hypothesis of Lemma 5.1 holds with $\alpha = 1/16 < 1/8$.

Invoke Fano. Lemma 5.1 with ρ as in Eq. (13), M from Eq. (11), and $\alpha = 1/16$ yields

$$\inf_{\widehat{f}} \sup_{\omega\in\mathcal{H}} P_{f_\omega}\Big(\Big\|\widehat{f} - f_\omega\Big\|_{L^2} \ge \rho\Big) \;\ge\; \frac{\sqrt{M}}{1+\sqrt{M}}\Big(1 - \tfrac{1}{8} - \sqrt{\tfrac{1/8}{\log M}}\Big).$$

For $m^d \ge 16$ (which we may assume by enlarging n_0), $\log M \ge (m^d/16)\log 2 \ge \log 2$, so $\sqrt{1/(8\log M)} \le \sqrt{1/(8\log 2)} \le 0.43$ and $\sqrt{M}/(1+\sqrt{M}) \ge 1/2$. Substituting,

$$\inf_{\widehat{f}} \sup_{\omega\in\mathcal{H}} P_{f_\omega}\Big(\Big\|\widehat{f} - f_\omega\Big\|_{L^2} \ge \rho\Big) \;\ge\; \tfrac{1}{2}\cdot\big(\tfrac{7}{8} - 0.43\big) \;\ge\; 0.22. \tag{15}$$

Setting $c_{\mathsf{prob}} := 0.22$ (or any smaller universal constant) gives Eq. (10) with $\rho = c_\rho m^{-s}$.

This completes the proof. □

6 Proof of the main theorem

Proof of Theorem 2.1. The proof proceeds in three steps. We first convert the probability lower bound of Lemma 5.2 into an expectation lower bound via Markov's inequality. We then choose m to balance the rate. We close by checking that all hypotheses of the lemmas are satisfied and assembling the constants.

Step 1 (probability to expectation via Markov). Fix an estimator $\widehat{f}$. For each $\omega \in \mathcal{H}$ define the event

$$\mathcal{E}_\omega \;:=\; \Big\{\Big\|\widehat{f} - f_\omega\Big\|_{L^2} \ge c_\rho\, m^{-s}\Big\}.$$

Markov's inequality applied to the non-negative random variable $\Big\|\widehat{f} - f_\omega\Big\|_{L^2}^2$ gives

$$\mathbb{E}_{P_{f_\omega}}\Big[\Big\|\widehat{f} - f_\omega\Big\|_{L^2}^2\Big] \;\ge\; (c_\rho\, m^{-s})^2 \cdot P_{f_\omega}(\mathcal{E}_\omega) \;=\; c_\rho^2\, m^{-2s} \cdot P_{f_\omega}(\mathcal{E}_\omega). \tag{16}$$

Taking the supremum over $\omega \in \mathcal{H} \subseteq \mathcal{F}$ (the inclusion holds by Lemma 3.1(i)) and then the infimum over estimators,

$$\inf_{\widehat{f}} \sup_{f^*\in\mathcal{F}} \mathbb{E}_{\mathcal{D}_n}\Big[\Big\|\widehat{f} - f^*\Big\|_{L^2}^2\Big] \;\ge\; \inf_{\widehat{f}} \sup_{\omega\in\mathcal{H}} \mathbb{E}_{P_{f_\omega}}\Big[\Big\|\widehat{f} - f_\omega\Big\|_{L^2}^2\Big] \;\ge\; c_\rho^2\, m^{-2s} \cdot \inf_{\widehat{f}} \sup_{\omega\in\mathcal{H}} P_{f_\omega}(\mathcal{E}_\omega). \tag{17}$$

The first inequality uses $\mathcal{H} \subseteq \mathcal{F}$; the second uses Eq. (16) on a per-ω basis, followed by the elementary identity $\inf_{\widehat{f}} \sup_\omega [c \cdot a_\omega(\widehat{f})] = c \cdot \inf_{\widehat{f}} \sup_\omega a_\omega(\widehat{f})$ for $c \ge 0$.

Step 2 (choose m to satisfy hypotheses, apply Lemma 5.2). We choose

$$m \;=\; m(n) \;:=\; \Big\lceil \big(n/(\alpha_\star\,\sigma^2)\big)^{1/(2s+d)} \Big\rceil, \tag{18}$$

where $\alpha_\star = (\log 2)/(256\,C_{\mathsf{KL}})$ is the constant from Lemma 5.2. We verify hypothesis Eq. (9): by construction $m \geq (n/(\alpha_\star\sigma^2))^{1/(2s+d)}$, hence $m^{2s+d} \geq n/(\alpha_\star\sigma^2)$, which rearranges to $n/(\sigma^2 m^{2s+d}) \leq \alpha_\star$, as required.

For Lemma 5.2 to apply we also need $m \geq 2$ and $m^d \geq 16$. Both hold for $n \geq n_0$ for some $n_0 = n_0(s, d, \sigma, \psi)$ (since $m(n) \to \infty$ as $n \to \infty$). We take n_0 to be the smallest such n.

Therefore by Lemma 5.2,

$$\inf_{\widehat{f}} \sup_{\omega\in\mathcal{H}} P_{f_\omega}(\mathcal{E}_\omega) \;\geq\; c_{\mathsf{prob}}. \tag{19}$$

Step 3 (assemble). Substituting Eq. (19) into Eq. (17),

$$\mathfrak{M}_n \;\geq\; c_\rho^2\, c_{\mathsf{prob}} \cdot m^{-2s}.$$

By Eq. (18) and the elementary inequality $\lceil x \rceil \leq x + 1 \leq 2x$ valid for $x \geq 1$,

$$m \;\leq\; 2\,\big(n/(\alpha_\star\,\sigma^2)\big)^{1/(2s+d)} \qquad \text{for } n \geq n_0, \tag{20}$$

where the bound $n/(\alpha_\star\sigma^2) \geq 1$ is absorbed into n_0. Therefore

$$m^{-2s} \;\geq\; 2^{-2s}\,\big(n/(\alpha_\star\sigma^2)\big)^{-2s/(2s+d)} \;=\; 2^{-2s}\,(\alpha_\star\sigma^2)^{2s/(2s+d)}\,n^{-2s/(2s+d)}. \tag{21}$$

Combining,

$$\mathfrak{M}_n \;\geq\; c_\rho^2\, c_{\mathsf{prob}}\, 2^{-2s}\,(\alpha_\star\sigma^2)^{2s/(2s+d)} \cdot n^{-2s/(2s+d)} \;=\; c_{\mathsf{lb}} \cdot n^{-2s/(2s+d)},$$

where

$$c_{\mathsf{lb}} \;:=\; c_\rho^2\, c_{\mathsf{prob}}\, 2^{-2s}\,(\alpha_\star\sigma^2)^{2s/(2s+d)} \;>\; 0.$$

This is Eq. (1), completing the proof. □

Remark 6.1. The constant c_{lb} depends on s, d, σ, and on the fixed bump function ψ (through $c_\psi, \|\psi\|_{L^2}, \|\psi\|_{L^\infty}, \|\psi\|_{W_2^s}, C_{\mathsf{KL}}$). The dependence on σ is $c_{\mathsf{lb}} \asymp (\sigma^2)^{2s/(2s+d)} = \sigma^{4s/(2s+d)}$, i.e., $c_{\mathsf{lb}} \to 0$ as $\sigma \to 0$, recovering the standard noise scaling for the minimax rate.

Remark 6.2 (Alternative reductions). Several variants of the Fano-style argument exist (e.g. Birgé–Massart's L_1-version, Yu's lemma with χ^2). All yield the same rate; we chose Tsybakov's formulation for transparency in the constant tracking.

References

[1] A. B. Tsybakov. *Introduction to Nonparametric Estimation.* Springer Series in Statistics. Springer, 2009.